

BUSINESS ANALYTICS CASE STUDY

TELECO CUSTOMER CHURN ANALYSIS REPORT

Introduction

Customer retention is a critical challenge in the telecommunications industry, where customer acquisition costs are significantly higher than customer retention costs. This project analyzes customer churn behavior using the Telco Customer Churn dataset to identify factors influencing customer attrition and provide actionable recommendations that can improve customer retention and business performance.

Business Problem Statement

The telecommunications company is experiencing customer churn, resulting in revenue loss and reduced customer lifetime value. The primary objective of this analysis is to identify the factors contributing to customer churn and determine which customer segments are at the highest risk of leaving.

Key business questions include:

- What percentage of customers are leaving the company?
- Which customer groups have the highest churn rates?
- What factors most influence customer churn?
- What actions can the company take to improve customer retention?

Dataset Description

The Telco Customer Churn dataset contains 7,043 customer records and includes demographic information, service subscriptions, contract details, billing information, and churn status.

Key variables analyzed include:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Internet Service
- Online Security
- Tech Support
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn

Additional calculated fields were created to support analysis, including tenure groups, churn rate measures, and numerical variables used for correlation analysis.

Data Analysis Process

The analysis followed the following steps:

Data Cleaning And Preparation

- Verified data quality and column data types in Power Query.
- Created tenure groups for customer segmentation.

- Created monthly charge bins for charge distribution analysis.
- Converted selected Yes/No fields into numerical values for correlation analysis.

Exploratory Data Analysis

- Calculated key performance indicators including total customers, active customers, churned customers, and churn rate.
- Examined churn distribution across customer segments.
- Analyzed churn by contract type, tenure, internet service, and senior citizen status.
- Investigated relationships between monthly charges and churn behavior.

Correlation Analysis

A correlation heatmap was developed to identify relationships between churn and key customer attributes including tenure, monthly charges, online security, tech support, partner status, and total charges.

Key Findings

- **Overall Churn Rate**

The company has a churn rate of 26.54%, representing 1,869 customers who have left the company.

- **Contract Type Impact**

Month-to-month customers exhibit significantly higher churn compared to customers on one-year and two-year contracts.

- **Tenure Impact**

Customer churn decreases as tenure increases. Customers within their first year exhibit the highest churn rate, while customers with more than 49 months of tenure demonstrate the lowest churn rate.

- **Internet Service Impact**

Fiber optic customers experience the highest churn rate at 41.89%, compared to DSL customers at 18.96% and customers without internet service at 7.40%.

- **Monthly Charges Impact**

Customers who churn have a higher average monthly charge (\$74.44) compared to customers who remain with the company (\$61.27), suggesting pricing may influence customer decisions.

- **Correlation Analysis**

The strongest negative relationship with churn is tenure (-0.35), indicating that longer-tenure customers are less likely to leave. Online Security and Tech Support also show negative relationships with churn, suggesting these services improve customer retention.

Business Insights

Several important customer behaviors emerged from the analysis:

- New customers are significantly more vulnerable to churn than long-term customers.
- Customers paying higher monthly charges are more likely to leave.
- Fiber optic customers represent the highest-risk service segment.
- Customers with Online Security and Tech Support services demonstrate stronger retention.
- Customers with partners and dependents tend to remain with the company longer and exhibit lower churn behavior.

These findings indicate that service value, pricing perception, and customer engagement play critical roles in customer retention.

RECOMMENDATIONS

1. **Strengthen New Customer Retention Programs:** Implement onboarding campaigns, welcome incentives, and proactive customer engagement during the first year of service.
2. **Encourage Long-Term Contracts:** Provide discounts and loyalty benefits for customers who transition from month-to-month contracts to annual agreements.
3. **Review Fiber Optic Service Experience:** Investigate pricing, service quality, and customer satisfaction among fiber optic customers to address the high churn rate.
4. **Promote Value-Added Services:** Increase adoption of Online Security and Tech Support through bundled packages and promotional offers.
5. **Evaluate Pricing Strategies:** Review pricing structures for high monthly charge customers and introduce personalized retention offers where appropriate.

CONCLUSION

This analysis identified tenure, monthly charges, internet service type, and value-added services as major drivers of customer churn. Customers with shorter tenure, higher monthly charges, and fiber optic subscriptions demonstrate the highest churn risk. By improving customer onboarding, promoting long-term contracts, enhancing service quality, and expanding value-added services, the company can significantly improve customer retention and reduce future churn.

Benedicta Izegbue
benedictaizegbue6@gmail.com